

BOK-GRPO

Best-of-K Group Relative Policy Optimization

Interleaved Point-to-Count for VLM Object Counting
Collapsing pass@K capability into pass@1

The pass@K Gap Problem

Metric	Value	Implication
pass@1	79.2%	Current single-attempt accuracy
pass@32	95.2%	Latent capability with 32 attempts
Gap	16.0pp	Untapped potential to recover

Standard GRPO ✗

- $A = (r - \mu) / \sigma \rightarrow$ vanishing gradients when $\sigma \rightarrow 0$
- 14/16 rollouts correct $\rightarrow$ no learning signal
- Cannot differentiate among high-reward paths

BOK-GRPO ✓

- $A = (\text{softmax}(z/\tau) - 1/K) \times K$
- Always separates best from rest, even when all good
- τ annealing: explore broadly $\rightarrow$ focus on single best

Interleaved Point-to-Count Paradigm

Turn	Input	Model Output	Environment
1	Original image + question	<point>{x,y, count:1}</point>	Draw red dot #1
2	Image + 1 red dot	<point>{x,y, count:2}</point>	Draw red dot #2
⋮	Image + (k-1) dots	<point>{x,y,	Draw red dot #k
Why Interleaved? (vs Single-Turn All-at-Once) <i>Humans count by pointing one-by-one. Sequential pointing creates explicit attention shift: each red dot serves as visual memory, preventing double-counting and attention drift.</i>			
Aspect	Single-Turn	Interleaved (Ours)	Why Better
Attention	Must track all at once	One object per turn	Reduces cognitive load
Feedback	No mid-process signal	Red dot = visual memory	Prevents re-counting
Error mode	Catastrophic all-or-nothing	Graceful per-step error	Error isolation

Core Algorithm: Softmax Advantage

Standard GRPO → Fails on Easy Groups

$$A_i = (r_i - \mu) / \sigma \Rightarrow \sigma \rightarrow 0 \text{ when } 14/16 \text{ correct}$$

BOK-GRPO: Softmax-Temperature Advantage

$$w_i = \text{softmax}(z_i / \tau) \quad A_i = (w_i - 1/K) \times K \quad \text{where } z_i = (r_i - \mu) / \sigma$$

τ Value	Behavior	Why This Range
$\tau \rightarrow 0$	Winner-take-all (pure BoK)	Maximum pressure, potentially unstable
$\tau = 0.3\text{--}0.7$	Top-K concentration	Practical: strong signal, still stable
$\tau \rightarrow \infty$	Uniform ($\approx$ std GRPO)	No selection pressure

Cosine Annealing: Why Start at $\tau=0.5$?

$$\tau(t) = \tau_f + (\tau_0 - \tau_f) \times \frac{1}{2} \times (1 + \cos(\pi t/T)) \quad [0.5 \rightarrow 0.3]$$

Why $\tau_{\text{init}}=0.5$, $\tau_{\text{final}}=0.3$?

Early training: policy is noisy, broader credit distribution ($\tau=0.5$) avoids premature convergence. Late training: policy is refined, sharper selection ($\tau=0.3$) focuses on the single best path.

4-Way Adaptive Routing

Route	Condition	Advantage	Why This Route
1. LowVar	$\sigma_t \leq \varepsilon$	$r - \text{batch_mean}$	Uniform group: use batch baseline
2. AllCorrect	All K correct	$A = 0$	Nothing to learn: skip
3.	pass > $\theta=0.50$	clip(z , ± 2.5)	Easy: z-norm suffices

Why $\theta_{\text{easy}} = 0.50$?

Empirically validated: below 0.50 too many groups incorrectly routed to EasyDrGRPO; above 0.50 too few groups get efficient z-norm gradients. 0.50 gives 84.7% effective gradient

4. BoK softmax	Default	$(50 \cdot \max(z, 0) - 50) / \sigma_K$	Hard: need BoK selection
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Why EasyDrGRPO Filter?

- When >50% rollouts succeed, z-norm gives valid gradients (enough variance)
- BoK softmax on easy groups wastes computation — z-norm is cheaper & sufficient
- Reserves BoK's concentrated selection for genuinely hard groups where it matters

LowVar	AllCorrect	EasyDrGRPO	BoK	Effective $\geq$
7%	15%	47%	31%	84.7%

Safety & Stability Mechanisms

Mechanism	Implementation	Why Needed
NaN Guard	<code>nan_to_num(scores, nan=0)</code>	Trajectory failures produce NaN rewards
Logit Cap	<code>clamp(logits, ±12.0)</code>	Extreme $z \rightarrow$ softmax overflow $\rightarrow$ NaN
τ -Adaptive Floor	$\tau_{\text{eff}} = \max(\tau, z_{\text{max}} /C)$	Prevents over-sharp softmax
Advantage Clip	<code>clamp(A, ±4.0)</code>	Bounds gradient magnitude
Collapse Detection	$\max(w) > 0.9 \rightarrow$ mix uniform	Single-traj domination $\rightarrow$ instability
$\text{eff_intensity} = \text{LR} \times \text{clip_high} \times \text{ppo_epochs}$ Sweet spot: $4\text{--}6e^{-7}$. V12 uses $1e^{-6} \times 0.28 \times 2 = 5.6e^{-7}$. Too high ($>1e^{-6}$) $\rightarrow$ NaN. Too low ($\leq 3e^{-7}$) $\rightarrow$ no learning.		
PPO Clip	<code>clip(r(t), 1-ε, 1+0.2ε)</code>	Trust region constraint

Dense Mask Point Reward

Per-Point Reward (3 Outcomes)

Outcome	Condition	Reward	Rationale
HIT (new)	Point in unused mask	1.0	Correct identification
HIT (dup)	Point in used mask	0.0	Penalize re-counting
MISS	Point outside all masks	decay(d)	Distance-based shaping

Distance Decay: Why 2-Phase Exponential?

Phase	Formula	Why This Design
Near ($d \leq 0.02$)	$\exp(-\alpha \cdot d)$, $\alpha=20$	Smooth gradient toward correct location
Far ($d > 0.02$)	$\exp(-\alpha \cdot d_f) \cdot \exp(-50(d - d_f))$	Steep drop: clearly wrong.

Why Masks Instead of GT Points?

RL training has NO per-step GT point annotations — only image + GT count. Pre-computed segmentation masks enable dense per-step reward without manual labeling. Mask-based matching handles objects of varying size (hit = inside contour, not just proximity).

Trajectory Reward Composition

$$R = 0.6 \times R_{\text{answer}} + 0.3 \times R_{\text{point}} + 0.1 \times R_{\text{format}}$$

Component	Weight	Design	Why This Weight
R_answer	0.6	Soft decay, cap=0.4	Primary goal is correct count
R_point	0.3	Dense mask per-step	Prevents reward sparsity
R_format	0.1	Binary compliance	Ensures parseable output

Answer Decay: Why Asymmetric Penalty?

Direction	k Factor	Why Asymmetric
Over-count (pred > GT)	k = 2.0	Hallucination is worse — counts non-existent objects
Under-count (pred < GT)	k = 1.0–1.5	Missing objects is less severe.

Why 0.6/0.3/0.1 and Not 1.0/0/0?

Pure answer reward is too sparse for multi-turn RL (one signal per trajectory). Dense point reward (0.3) provides per-step learning signal. Format (0.1) prevents structural degeneration.

Consistency Check

If $\text{pred_answer} \neq \text{\#pred_points}$: $R_{\text{answer}} \times = 0.5$

Training Configuration

Parameter	Value	Why This Choice
Base Model	Qwen2.5-VL-7B	Strong VLM with point/grounding capability
SFT Pre-train	30k / 3537 steps	Establish counting format before RL
K (rollouts)	16	Enough diversity for softmax advantage
Learning Rate	$1e^{-6}$	Part of eff_intensity sweet spot
PPO Epochs	2	Balance between update strength and stability
Clip Ratio	0.28	$\text{eff_intensity} = \text{LR} \times \text{clip} \times \text{ppo} = 5.6e^{-7}$
KL Coeff	0.03	Prevent drift while allowing adaptation
τ Schedule	0.5 $\rightarrow$ 0.3 cosine	Explore-then-exploit temperature
BOK_CLIP	4.0	Bound advantage magnitude
θ_{easy}	0.50	Validated optimal routing threshold
Max Turns	11	Cover GT ≤ 10 with margin
Reward	0.6/0.3/0.1	Answer + dense point + format

Data Strategy

Dataset Comparison

Dataset	Samples	Easy/Med/Hard	Use Case
0_10 (sparse only)	11,455	30/41/29%	V7-V12: sparse counting
hard_only	1,808	18/40/42%	V14: hard data focus
mixed (hard+easy)	5,792	17/32/51%	V14-16: balanced training

Why Mixed Data (Not Pure Hard)?

Pure hard data (V14-hard) showed: answer reward=0.74 but pixmo-test=79.4% (worse than V12). Easy examples provide gradient signal when hard examples are all-wrong. Mixed data maintains BoK routing diversity and prevents policy collapse.

Impact of Data Choice

Experiment	Data	pixmo-test	Finding
V12 (best)	0_10 sparse	82.04%	Best overall accuracy
V14-hard	hard_only	79.40%	Too narrow, overfits to hard
V14-mixed	mixed	80.34%	Better than hard-only
V15	mixed + low LR	80.15%	LR too conservative

Main Results

Benchmark Comparison (GT 2–10)

Model	pixmo-test	countbench	Method
SFT Base	75.6%	—	Supervised Only
V7 Std GRPO	80.72%	—	Standard GRPO
V14 Mixed	80.34%	78.21%	Mixed + BoK
V15 Low-LR	80.15%	79.02%	Conservative
V12 BOK-GRPO	82.04%	79.43%	BOK-GRPO (best)

V12 Detailed Breakdown

Split	Samples	Accuracy
Overall	529	82.04%
Easy (GT 0–5)	239	84.10%
Hard (GT 5+)	290	80.34%
With History	—	83.36%

Key Result

+6.44pp over SFT (75.6→82.04%) | +1.32pp over Standard GRPO (80.72→82.04%) |
pass@1–@32 gap: 19.6pp → ~13pp (–33%)

Ablation: What Matters

Design Choice	Without	With (Ours)	Why It Helps
Interleaved multi-turn	All-at-once	Sequential+feedback	Visual memory via red dots
Dense point reward	Answer-only	+mask point (0.3)	Per-step signal prevents sparsity
BoK softmax adv.	Z-norm GRPO	Softmax weighting	+1.32pp: learns from best paths
Mask matching	Distance-only	Hit/miss/duplicate	Size-aware spatial grounding
Consistency check	None	Drop 0.5 if $\Delta > 0.5$	Redlines
Most Impactful <i>Dense point reward (prevents reward hacking) and BoK softmax advantage (+1.32pp) are the two most critical design decisions.</i>			
Soft answer decay	Binary 0/1	exp decay, cap=0.4	Finer gradient for wrong answers
τ cosine annealing	Fixed $\tau=0.3$	0.5 $\rightarrow$ 0.3 cosine	Explore early, exploit late

Key Findings

Finding	Evidence	Implication
+6.44pp over SFT	75.6% → 82.04%	RL training is effective for counting
+1.32pp over Std GRPO	80.72% → 82.04%	BoK advantage > z-norm advantage
pass@1-@32 gap -33%	19.6pp → ~13pp	Capability collapse is working
Zero NaN events	eff_intensity=5.6e-7	Safety mechanisms are sufficient

Stability Metrics

Metric	Value	Status
96.6% format	Maintained throughout	Format reward preserves
eff_intensity	5.6e ⁻⁷	✓ In sweet spot (4-6e-7)
NaN Events	0	✓ Fully stable
Format Compliance	96.6%	✓ High

Future Directions

Performance Targets

Scenario	Current	Target	Approach
Sparse (≤ 10 obj)	82%	$\geq 95\%$	BOK-GRPO + curriculum
Dense (11–50 obj)	$\sim 30\%$	$\geq 50\%$	Dense scene + adaptive routing

Algorithm Extensions

- Per-turn process reward (step-level advantage)
- Adaptive routing thresholds (dynamic θ)
- History-efficient prompting
- Curriculum: easy $\rightarrow$ hard scheduling

Scaling & Transfer

- Qwen2.5-VL-72B with BOK-GRPO
- Cross-domain: detection, segmentation
- Real-time streaming visual feedback
- Multi-category object counting

Appendix: Algorithm Pseudocode

```
def BOK_GRP0(rollouts, K=16,  $\tau_{init}$ =0.5,  $\tau_{final}$ =0.3, T=90):
    scores = trajectory_rewards(rollouts)      #  $R = 0.6*ans + 0.3*pt + 0.1*fmt$ 
     $\tau = \tau_f + (\tau_0 - \tau_f) * 0.5 * (1 + \cos(\pi*t/T))$     # cosine annealing

    for group in rollouts.group_by_prompt():
         $\sigma$ , pass_rate = group.std(), fraction(score >  $\theta$ )

        if  $\sigma \leq \epsilon$ :                                # Route 1: LowVar
            A = score - batch_mean
        elif all(score >  $\theta$ ):                            # Route 2: AllCorrect
            A = 0
        elif pass_rate >  $\theta_{easy}$ :                        # Route 3: EasyDrGRP0
            A = clip((score -  $\mu$ )/ $\sigma$ ,  $\pm 2.5$ )
        else:                                                # Route 4: BoK Softmax
            z = (score -  $\mu$ ) /  $\sigma$ 
             $\tau_{eff} = \max(\tau, \max|z|/C, \tau_{min})$         # adaptive floor
            w = softmax(z /  $\tau_{eff}$ )
            w = (1- $\alpha$ )*w +  $\alpha/K$                         # uniform mixing
            A = (w - 1/K) * K                                # BoK advantage

        A = clamp(A, -BOK_CLIP, BOK_CLIP)                  # final clip

    # PPO update
    L = -E[min(r( $\theta$ )*A, clip(r( $\theta$ ),  $1 \pm \epsilon$ )*A)] +  $\beta * D_{KL}(\pi_{\theta} || \pi_{ref})$ 
```